

Faculty of Engineering

Internal Design Document

MQTT Protocol Design for Distributed Real-Time Telemetry System for QUT Microgrid

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Version History

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1 Overview

This document describes the MQTT protocol design for the distributed real-time telemetry system developed as part of the EGH490 thesis project at the Queensland University of Technology. The system acquires electrical measurments from STM32-based data acquisition nodes deployed across the QUT microgrid and transports them to centralised time-series historian via an MQTT message broker,

The design covers broker selection and configuration, topic heirachy, payload format, quality of service strategy, client architecture, security posture, and historian integration. All decisions are scoped to a two-node validation testbed operating over a wired ethernet LAN, with the architecture designed to generalise to an arbitrary number of nodes without structural change.

2 Scope

The following contraints apply to all documented design decisions.

Table 1: Design Constraints

Parameter	Value
Transport Medium	Wired Ethernet (LAN) only
Number of DAQ nodes (validation testbed)	2
DAQ node MCUJ	STM32 (IEEE 1588 hardware timestamping supported)
Time synchronisation protocol	PTP (IEEE 1588-2019)
Historian	InfluxDB v2
Target measurement standards	IEC 61000-4-30 Class S, IEEE 1459-2010

Wireless transport is specifically out of scope. The wired only constraint eliminates RF-induced jitter as a variable in timestamp accuracy, permitting PTP synchronisation to operate within the sub-microsectond range achievable on a switched Ethernet topology. The two-node testbed reflects hardware availabilityat the time of validation; the architecture imposes no upper bound on the number of nodes.

3 Broker Selection & Configuration

Eclipse Mosquitto is selected as the MQTT broker. This is suitable of a research testbed with two DAQ nodes as it is:

- lightweight
- well documented
- suitable for co-deployment on the historian server alongside InfluxDB without significant resource contention
- carries no licensing contraintes under the eclipse public licence

Alternatives such as EMQX and HiveMQ were considered but are designed for large-scale deployments with thousands of concurrent clients; the additional operational complexity is not justified at this scale.

The broker runs on the microgrid historian server, bound to the lab LAN interface. Key configuration parameters are as follows:

```
listener 1883 <lab_lan_ip>
allow_anonymous false
password_file /etc/mosquitto/passwd

persistence true
persistence_location /var/lib/mosquitto/

max_inflight_messages 100
max_queued_messages   1000

log_type error
log_type warning
log_type notice
```

Port 1884 is used for internal lab LAN traffic. TLS on port 8883 is reserved as a future hardening measure and discussed in section X. Persistence is enabled so that QoS 1 messages survive a broker restart without loss. The `max_inflight_messages` and `max_queued_messages` parameters are bounded to prevent unbounded memory growth under a backlogged subscriber.

4 Topic Hierachy

4.1 Naming Scheme

Topics follow a four-level hierachy:

```
microgrid/<node_id>/<subsystem>/<measurement>
```

Table 2: Hierachy Description

Segment	Description	Example Values
`node_id`	Fixes, zero-padded node identifier	`node_1`, `node_2`
`subsystem`	Electrical quantity category	`voltage`, `current`, `power`, `frequency`
`measurement`	Specific measured quality	`phase_a`, `active`, `reactive`, `fundamental`

Node identifiers are fixed strings assigned at firmware flash time. MAC addresses and IP addresses are explicitly avoided as node identifiers as both are mutable and would invalidate historical records if hardware is replaced. Measurement names are aligned with IEEE 1459-2010 terminology where applicable

4.2 Example Topics

4.3 Wildcard Subscription Patterns

Table 3: Wildcard Subscription Patterns

Pattern	Use Case
microgrid/#	full subscription: used by the InfluxDB historian bridge
microgrid//status/heartbeat	node liveness monitoring across all nodes
microgrid/node_01/#	All measurements from a single node (debugging)
microgrid//voltage#	All voltage measurements across all nodes

5 Payload Format

5.1 Format Selection

Payloads are serialised as UTF-8 JSON. This prioritises developer ergonomics and debuggability during the research phase: payloads are huiman-readable, trivially inspected with any MQTT client and require no schema negotiation between publisher and subscriber. Migration to a binary format such as MessagePack or CBOR is possible if bandwidth or parsing overhead becomes a concern, but is not anticipated on a wired LAN with two nodes.

5.2 Single-Sample Payload

```
{
  "ts": 1713400000.123456,
  "v": 231.45
}
```

Table 4: Single-Sample Payload Description

Field	Type	Description
`ts`	float64	PTP-synchronised Unix timestamp (fractional seconds) sourced from the STM32 IEEE 1588 hardware clock
`v`	float32	Measurement value in SI units

The timestamp is sourcef from the STM32 IEEE 1588 hardware timestamping peripheral, not the system tick counter. This preserves PTP synchronisation accurcy end–to-end, from the ADC sample through to the broker and into the InfluxDB point timestamp

Units are implied by the topic rather than included in the payload. For example, a message on `microgrid/node_01/volt` is understood to carry a value in volts.

5.3 Burst Payloads

For high-frequency acquisition where per-message overhead is a concern, multiple samples may be batched into a single publish:

5.4 Single-Sample Payload

```
{
  "ts": 1713400000.000000,
  "dt": 0.000312,
  "v":  [231.45, 231.47, 231.43, 231.50]
}
```

Table 5: Burst Payload Description

Field	Type	Description
ts	float64	Timestamp of the first sample in the batch
dt	float64	Inter-sample interval in seconds
v	float32	Ordered array of sample values

The historian bridge reconstructs individual timestamps as $t_s + i \cdot \Delta t$ for sample index i, and writes each as a separate InfluxDB point.

The decision between single–sample and burst sample publishing depends on the ADC sample rate (yet to be finalided). This should be resolved before the firmware publish loop is implemented.